

1: How many ways can 5 differently colored balls be placed into bins labeled 1–8 so that no bin contains more than one ball?

Solution: We first choose which 5 of the 8 bins will be occupied and then multiply by the number of different ways to arrange the 5 balls:

$$\binom{8}{5} \cdot 5! = \boxed{6720}$$

2: Thirty items are arranged in a 6-by-5 array as shown below:

A_1	A_2	A_3	A_4	A_5
A_6	A_7	A_8	A_9	A_{10}
A_{11}	A_{12}	A_{13}	A_{14}	A_{15}
A_{16}	A_{17}	A_{18}	A_{19}	A_{20}
A_{21}	A_{22}	A_{23}	A_{24}	A_{25}
A_{26}	A_{27}	A_{28}	A_{29}	A_{30}

Calculate the number of ways to form a set of three distinct items such that no two of the selected items are in the same row or column.

Solution: We first choose the row, then the associated column, and then account for the different ways to pair them together:

$$\binom{5}{3} \cdot \binom{6}{3} \cdot 3! = \frac{5!}{3!(5-3)!} \cdot \frac{6!}{3!(6-3)!} \cdot 3! = \boxed{1200}$$